MAT 1272 Statistics

Chapter 5: Discrete Random Variables and Probability Distributions

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Chapter 5 Roadmap

- 5.1 Random variables
- 5.2 Probability distributions
- 5.3 Mean and standard deviation
- 5.4 Binomial probability distribution
- 5.5 Hypergeometric probability distribution

Big Idea

A probability distribution describes the possible values of a random variable and how likely those values are.

5.1 Random Variables

Definition

A random variable is a variable whose value depends on the outcome of a random experiment.

Examples

- Number of heads in three coin tosses
- Number of students absent today
- Time required to complete an exam
- Height of a randomly selected student

Discrete Random Variables

Definition

A discrete random variable assumes countable values.

Examples

- Number of children in a family
- Number of emails received today
- Number of cars in a parking lot
- Number of computers repaired in a day

Key Idea

Discrete means countable.

Continuous Random Variables

Definition

A continuous random variable can assume any value in an interval.

Examples

- Height
- Weight
- Time
- Temperature
- Amount of liquid in a bottle

Key Idea

Continuous means measurable.

Quick Check

Classify each random variable as discrete or continuous.

1. Number of books in a backpack
2. Time spent studying
3. Number of customers entering a store
4. Weight of a newborn baby
5. Number of text messages received today

Question

Can we count it, or do we measure it?

Definition

A probability distribution of a discrete random variable lists all possible values of the random variable and their corresponding probabilities.

x	0	1	2
$P(x)$	0.25	0.50	0.25

Example

If two coins are tossed, x can represent the number of heads.

Two Conditions

A valid probability distribution must satisfy

$$0 \leq P(x) \leq 1$$

for every value of x, and

$$\sum P(x) = 1.$$

Meaning

Each probability must be between 0 and 1, and all probabilities together must add to 1.

Example: Is This Valid?

x	0	1	2	3
$P(x)$	0.10	0.30	0.40	0.20

$$0.10 + 0.30 + 0.40 + 0.20 = 1.00$$

Conclusion

This is a valid probability distribution.

Example: Not Valid

x	0	1	2	3
$P(x)$	0.08	0.11	0.39	0.27

$$0.08 + 0.11 + 0.39 + 0.27 = 0.85$$

Conclusion

This is not valid because the probabilities do not add to 1.

Finding Probabilities from a Table

Suppose

x	0	1	2	3
$P(x)$	0.15	0.20	0.35	0.30

Find:

$$P(X = 2), \qquad P(X \geq 2), \qquad P(X \leq 1)$$

$$P(X = 2) = 0.35$$

$$P(X \geq 2) = 0.35 + 0.30 = 0.65$$

$$P(X \leq 1) = 0.15 + 0.20 = 0.35$$

5.3 Mean of a Discrete Random Variable

Mean or Expected Value

$$\mu = E(X) = \sum xP(x)$$

Interpretation

The mean is the long-run average value of the random variable after many repetitions.

Example: Mean

x	$P(x)$	$xP(x)$
0	0.15	0.00
1	0.20	0.20
2	0.35	0.70
3	0.30	0.90
		1.80

$$\mu = \sum xP(x) = 1.80$$

Meaning

In the long run, we expect an average of 1.80 occurrences per trial.

Standard Deviation

Shortcut Formula

$$\sigma = \sqrt{\sum x^2 P(x) - \mu^2}$$

Interpretation

The standard deviation measures the spread of the values of x around the mean.

Example: Standard Deviation

x	$P(x)$	$xP(x)$	$x^2P(x)$
0	.15	.00	.00
1	.20	.20	.20
2	.35	.70	1.40
3	.30	.90	2.70
		1.80	4.30

$$\sigma = \sqrt{4.30 - (1.80)^2} = \sqrt{1.06} \approx 1.03$$

TI–84: Mean and Standard Deviation

1. Press STAT.
2. Choose EDIT.
3. Enter the x-values into L1.
4. Enter the probabilities into L2.
5. Press STAT.
6. Move to CALC.
7. Choose 1-Var Stats.
8. Type:

   ```
   1-Var Stats L1,L2
   ```

9. Press ENTER.

TI–84 Output

For a probability distribution:

$$\bar{x}$$

gives the mean

$$\mu.$$

The calculator also gives

$$\sigma_x,$$

which is the standard deviation.

Important

For probability distributions, use σ_x, not S_x.

5.4 Binomial Probability Distribution

Binomial Experiment

A binomial experiment has:

1. A fixed number of trials n.
2. Only two outcomes: success or failure.
3. The probability of success p stays the same.
4. The trials are independent.

Success and Failure

Notation

$$p = P(\text{success})$$

$$q = P(\text{failure}) = 1 - p$$

Example

If the probability of success is

$$p = 0.30,$$

then

$$q = 1 - 0.30 = 0.70.$$

Binomial Random Variable

Definition

A binomial random variable counts the number of successes in n trials.

Examples

- Number of heads in 10 coin tosses
- Number of students who pass an exam
- Number of defective items in a sample
- Number of adults who say yes in a survey

Binomial Formula

Formula

$$P(x) = \binom{n}{x} p^x q^{n-x}$$

where

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}.$$

Meaning

This gives the probability of exactly x successes in n trials.

Example: Exactly 3 Heads

A coin is tossed 5 times. Find the probability of exactly 3 heads.

$$n = 5, \qquad x = 3, \qquad p = 0.5, \qquad q = 0.5$$

$$P(3) = \binom{5}{3}(0.5)^3(0.5)^2$$

$$P(3) = 10(0.125)(0.25) = 0.3125$$

$$\boxed{0.3125}$$

Mean and Standard Deviation of a Binomial Variable

Formulas

$$\mu = np$$

$$\sigma = \sqrt{npq}$$

Example

If $n = 20$, $p = 0.40$, and $q = 0.60$, then

$$\mu = (20)(0.40) = 8$$

$$\sigma = \sqrt{(20)(0.40)(0.60)} \approx 2.19.$$

TI–84: Binomial Probability

Exactly x successes

Use

```
binompdf(n,p,x)
```

Example

For exactly 3 heads in 5 tosses:

```
binompdf(5,.5,3)
```

The answer is

$$0.3125.$$

TI–84: Cumulative Binomial Probability

At most x successes

Use

```
binomcdf(n,p,x)
```

This computes

$$P(X \leq x).$$

Example

```
binomcdf(5,.5,3)
```

computes

$$P(X \leq 3).$$

At Least x Successes

Important

For at least x successes:

$$P(X \geq x) = 1 - P(X \leq x - 1).$$

Use

$$1 - \texttt{binomcdf(n,p,x-1)}.$$

Example

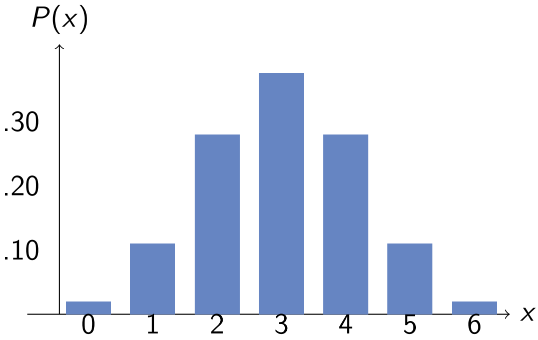
$$P(X \geq 4) = 1 - P(X \leq 3).$$

Visualizing a Binomial Distribution

Why this matters

The binomial distribution is discrete, but its histogram can begin to look like a bell-shaped curve.
This prepares us for the normal distribution in Chapter 6.

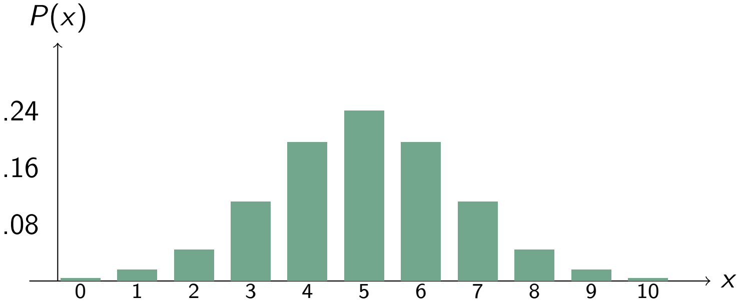
Binomial Histogram: $n = 6,\ p = 0.5$



Observation

The distribution is symmetric and mound-shaped.

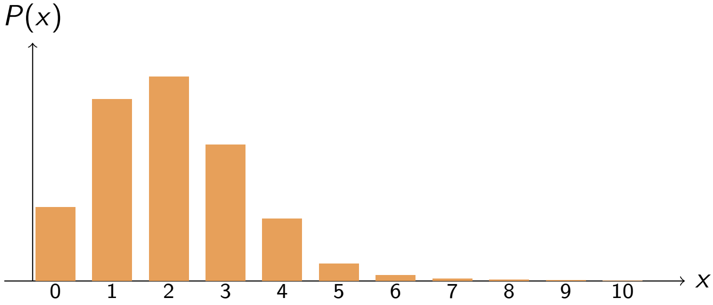
Binomial Histogram: $n = 10,\ p = 0.5$



Observation

As n increases, the shape becomes smoother and more bell-shaped.

Binomial Histogram: $n = 10,\ p = 0.2$



Observation

When p is far from 0.5, the distribution may be skewed.

Connection to the Normal Distribution

Big Picture

When n is large and p is not too close to 0 or 1, the binomial distribution often becomes approximately bell-shaped.

Preview

In Chapter 6, we study the normal distribution, the smooth bell-shaped curve that appears often in statistics.

5.5 Hypergeometric Distribution

When do we use it?

The hypergeometric distribution is used when sampling is done without replacement from a finite population.

Key Difference

Binomial: probability stays constant.
Hypergeometric: probability changes because items are not replaced.

Binomial vs. Hypergeometric

Binomial	Hypergeometric
With replacement	Without replacement
Independent trials	Dependent trials
p stays constant	p changes
Uses n, p, x	Uses N, r, n, x

Hypergeometric Formula

Formula

$$P(x) = \frac{\binom{r}{x}\binom{N-r}{n-x}}{\binom{N}{n}}$$

where

$$N = \text{population size}$$

$$r = \text{number of successes in the population}$$

$$n = \text{sample size}$$

$$x = \text{number of successes in the sample}$$

Example: Auto Parts

A shipment contains 25 parts.

- 20 are good.
- 5 are defective.
- 4 parts are selected without replacement.

Find the probability that exactly 3 are good.

$$N = 25, \qquad r = 20, \qquad n = 4, \qquad x = 3$$

Hypergeometric Computation

$$P(3) = \frac{\binom{20}{3}\binom{5}{1}}{\binom{25}{4}}$$

$$P(3) \approx 0.4506$$

Interpretation

There is about a 45.06% chance that exactly 3 of the 4 selected parts are good.

Chapter 5 Summary

- Random variables can be discrete or continuous.
- A probability distribution lists values and probabilities.
- A valid probability distribution has probabilities that add to 1.
- Mean:

$$\mu = \sum xP(x)$$

- Standard deviation:

$$\sigma = \sqrt{\sum x^2 P(x) - \mu^2}$$

Chapter 5 Summary Continued

- Binomial experiments have fixed n, two outcomes, constant p, and independent trials.
- Binomial probability:

$$P(x) = \binom{n}{x} p^x q^{n-x}$$

- Binomial mean:

$$\mu = np$$

- Binomial standard deviation:

$$\sigma = \sqrt{npq}$$

- Hypergeometric applies when sampling is without replacement.

Group Activity

1. Classify random variables as discrete or continuous.
2. Verify whether a table is a valid probability distribution.
3. Compute the mean and standard deviation from a table.
4. Decide whether a situation is binomial.
5. Use binompdf and binomcdf.
6. Decide whether a situation is hypergeometric.

Questions?